Luiz F. G. dos Santos, Ph.D.

Machine Learning Engineer | LLM Systems, Applied AI, and Scalable ML

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Publications: https://bit.ly/2TV4aik

Professional Summary

– Machine learning engineer and AI research lead with a Ph.D. in Physics and hands-on experience taking LLM, transformer, and applied AI systems from research prototypes through production inference.
– Focus areas include distributed training, high-throughput data pipelines, GPU performance, mixed precision, model evaluation, and reliable serving with PyTorch, Triton, TensorRT, Kubernetes, and MLflow.
– Guides ambiguous ML initiatives with engineers, scientists, product stakeholders, and business partners.

Skills

Core ML: LLM Systems, Transformers, Tokenization, Representation Learning, RAG, Natural Language Processing, Computer Vision, Applied AI, Model Evaluation

Training and GPU: Distributed Training, Training Optimization, PyTorch, CUDA, GPU Acceleration, Mixed Precision, FlashAttention, Composer, Accelerate

Inference and Deployment: Inference Optimization, TensorRT-LLM, TensorRT, Triton, Kubernetes, MLflow, Hugging Face, Docker

Data and Configuration: High-Throughput Data, GCS, MDS, Hydra, Pydantic, Python

Leadership: Technical Leadership, Cross-functional Collaboration, Roadmap Ownership, Performance Engineering

Professional Experience

Delphos Labs Inc. | Houston, TX | Jan 2025 – Present

AI Researcher Lead | Nov 2025 – Present

Lead AI research and ML systems engineering for applied AI products, focused on tokenizer development, learned technical representations, distributed training infrastructure, evaluation, and production inference.

– Directed core development of an ML training platform, spanning distributed pretraining and fine-tuning, Hydra/Pydantic experiment configuration, MLflow-backed model management, and objective model evaluation.
– Productionized model applications for embedding, naming, and detection use cases through Triton, TensorRT, TensorRT-LLM, Kubernetes, and MLflow Model Registry.
– Improved GPU training reliability and throughput with Composer/Accelerate, FlashAttention 3, mixed precision controls, distributed data-loading fixes, and preflight checks.
– Built evaluation, reporting, and autonomous experiment workflows that run time-budgeted training jobs and keep or reject changes based on objective metrics, reliability, and performance tradeoffs.

AI Researcher | Jan 2025 – Oct 2025

– Built high-throughput tokenization and dataset workflows for technical code-like representations, including custom WordPiece tokenizer training, large-scale dataset tokenization, GCS/MDS datasets, and repeatable transformer pretraining pipelines.
– Trained and evaluated ModernBERT/FlexBERT-style embedding models over large technical corpora, comparing model quality, throughput, and deployment readiness through repeatable experiment templates.

Shell | Houston, TX | Mar 2022 – Dec 2024

Principal Data Scientist and AI Researcher | Enterprise GenAI and applied AI systems

Served as Principal Data Scientist leading LLM, RAG, NLP, and automation projects from prototype through enterprise adoption.

– Deployed LLM-based agents for file analysis, information extraction, and analytical automation, turning ambiguous business needs into adopted AI capabilities.

– Designed RAG systems that improved contextual grounding for enterprise LLM workflows and reduced dependence on manual document review.
– Directed NLP and deep learning projects for unstructured-data extraction, structuring, and analysis across technical and business domains.
– Built predictive, optimization, sequence-modeling, and reinforcement-learning models, balancing scientific rigor, performance, interpretability, and deployment readiness.
– Partnered with business stakeholders, technology teams, and external collaborators to define AI roadmaps, communicate tradeoffs, and move prototypes into production use.

CIRES / University of Colorado Boulder | Boulder, CO | May 2021 – Mar 2022

Postdoctoral Associate | AI for space sciences

Built deep learning and statistical models for solar-wind time series, remote-sensing imagery, and sensor data, combining computer vision, signal processing, and uncertainty quantification.

– Extracted supervised and unsupervised features from extreme ultraviolet (EUV) solar images across multiple frequencies and from magnetic-field data for deep learning forecasting models.
– Developed quantitative uncertainty models for machine learning algorithms in space-weather forecasting, improving reliability, risk interpretation, and communication of model limitations.

Frontier Development Lab | Mountain View, CA | Jun 2019 – May 2021

Researcher | Expanding the capabilities of NASA's Solar Dynamics Observatory

– Developed a calibration technique for EUV instruments using a Convolutional Neural Network (CNN) over morphological and brightness features from AIA observations, improving image-data quality and operational efficiency.
– Used unsupervised deep learning on multi-channel Atmospheric Imaging Assembly (AIA) data, using U-Nets and Generative Adversarial Networks (GANs) for image-to-image translation and synthetic solar-corona imagery.

NASA Goddard Space Flight Center | Washington Metro, DC | Jan 2017 – May 2021

Research Assistant | Machine learning for magnetic-field time series, mission data, and remote imagery

– Reduced, calibrated, and analyzed SDO, STEREO, and WIND mission data; used machine learning on in situ time series and remote image data while maintaining data quality for downstream modeling.
– Used scripting, compiled programming, vectorized workflows, parallel processing, GPU programming, and Bayesian/frequentist statistics to improve computational efficiency, robustness, and interpretation.
– Coordinated collaborative projects, mentored students, and delivered presentations that communicated complex model results to technical and non-technical audiences.

Education

Ph.D. in Physics, The Catholic University of America, Washington, DC | Sep 2016 – Apr 2021

Graduated with distinction; GPA: 3.62/4.0

Master of Science in Physics, The Catholic University of America, Washington, DC | Sep 2017 – May 2018

GPA: 3.57/4.0

Bachelor in Physics, University of Brasilia / St. John's University | Sep 2010 – Jul 2015

Merit scholarship; GPA: 3.92/4.0

Honors and Publications

John C. Mather Nobel Scholar and NASA Frontier Development Lab / SETI Institute Award of Merit. Selected peer-reviewed publications and conference papers in machine learning, remote sensing, space-weather forecasting, deep learning, and reinforcement learning. Full publication list: `https://bit.ly/2TV4aik`